

Dataset Shortlist for MED-VISION / MedChron AI

1. Objective of Dataset Selection

The goal of dataset selection is to finalize a medical imaging dataset that supports the following project requirements:

- Disease classification using **VGG16 + transfer learning + fine-tuning**
- Image preprocessing and enhancement
- Morphological operations and image segmentation
- ROI (Region of Interest) extraction where feasible
- Explainability (Grad-CAM / localization overlays)
- Future extension toward **patient timeline / disease progression / digital twin**

The dataset should therefore be evaluated not only for classification labels, but also for:

- imaging modality
- annotation quality
- ROI feasibility
- preprocessing suitability
- model training practicality
- long-term project scalability

2. Recommended Dataset Categories

The shortlisted datasets are grouped into the following categories:

1. **Chest X-ray datasets**
2. **Chest X-ray ROI / lung segmentation datasets**
3. **Breast cancer / mammography datasets**
4. **Brain tumor / MRI datasets**
5. **Lung cancer / CT datasets**
6. **Histopathology / tissue cancer datasets**

3. Dataset Shortlist

A. Chest X-ray Datasets

(Best overall category for the current MED-VISION / MedChron AI direction)

These are the strongest options if the project is built around **chest radiology, disease classification, ROI extraction, and a professional web demo.**

A1. VinDr-CXR

Best for: classification + abnormality localization + strong professional project scope

Modality

- Chest X-ray (CXR)

Why this dataset is important

- One of the strongest public chest X-ray datasets for AI research
- Suitable for **classification + localization**
- Supports a more professional medical imaging pipeline compared to simple image-label datasets

Key features

- **18,000 chest X-rays**
- **22 local findings**
- **6 global disease labels**
- Radiologist-provided annotations
- Bounding-box based abnormality localization
- DICOM-format medical images

Why it is useful for our project

- Strong dataset for **VGG16 fine-tuning**
- Good for **preprocessing and enhancement experiments**
- Useful for **abnormality-aware ROI strategy**
- Can support Grad-CAM + annotation overlay based presentation
- Good foundation for a chest-radiology intelligence platform

Links

- Official VinDr dataset page: <https://vindr.ai/dataset-cxr-backup>
- PhysioNet release: <https://physionet.org/content/vindr-cxr/1.0.0/>

- Dataset paper: <https://www.nature.com/articles/s41597-022-01498-w>

A2. RSNA Pneumonia Detection Challenge

Best for: fast MVP + pneumonia detection + ROI via bounding boxes

Modality

- Chest X-ray

Why this dataset is important

- Strong option if the project wants a **clean first implementation** with clear disease labels and ROI information
- Easier to operationalize compared to more complex multi-label datasets

Key features

- Chest X-ray dataset focused on pneumonia detection
- Includes **pneumonia labels**
- Includes **bounding boxes for suspicious lung opacities**
- Good for ROI-based preprocessing and disease localization

Why it is useful for our project

- Very suitable for:
 - preprocessing pipeline
 - segmentation experiments
 - ROI extraction
 - classification + localization demo
- Easier to turn into a polished first web prototype

Links

- Official RSNA challenge page: <https://www.rsna.org/artificial-intelligence/ai-image-challenge/rsna-pneumonia-detection-challenge-2018>
- Kaggle mirror: <https://www.kaggle.com/datasets/parin30/rsna-pneumonia-detection>

A3. NIH ChestX-ray14

Best for: large-scale chest disease classification

Modality

- Chest X-ray

Key features

- **112,000+ chest X-rays**
- Multiple thoracic disease labels
- Widely used benchmark dataset

Why it is useful

- Good if the goal is large-scale classification benchmarking
- Strong for transfer learning experiments
- Can support disease classification research

Limitation for current project

- ROI / segmentation support is not as straightforward as datasets that provide cleaner localization support

Links

- NIH/Kaggle dataset mirror: <https://www.kaggle.com/datasets/nih-chest-xrays/data>

A4. CheXpert

Best for: strong research benchmark in chest radiology classification

Modality

- Chest X-ray

Key features

- Large chest radiograph dataset
- Multi-label thoracic disease prediction
- Created by Stanford ML Group

Why it is useful

- Strong research credibility
- Good for classification benchmarking
- Useful for transfer learning and multi-label disease prediction

Link

- Official dataset page: <https://stanfordmlgroup.github.io/competitions/chexpert/>

A5. MIMIC-CXR

Best for: advanced research / report generation / large-scale radiology data

Modality

- Chest X-ray

Key features

- Very large chest radiograph dataset
- Includes associated radiology reports
- Strong dataset for advanced research workflows

Why it is useful

- Helpful if the project later extends into:
 - report generation
 - radiology text integration
 - multimodal AI
- Strong long-term research dataset

Link

- Official dataset page: <https://physionet.org/content/mimic-cxr/2.1.0/>

B. Chest X-ray ROI / Lung Segmentation Datasets

(Recommended as ROI-support datasets for chest X-ray projects)

These datasets are especially useful if we choose a **chest X-ray main dataset** and want to add **lung ROI extraction / segmentation / morphology** before classification.

B1. Montgomery + Shenzhen Lung Segmentation Collections

Best for: lung ROI extraction and segmentation support

Modality

- Chest X-ray

Why these datasets are important

- Frequently used in lung segmentation research
- Useful for building lung-field ROI extraction
- Good for testing:
 - thresholding
 - morphology
 - contour extraction
 - segmentation pipelines

Key features

- Chest X-rays with **lung segmentation masks**
- Suitable for ROI-based preprocessing and segmentation model prototyping
- Can support lung-field cropping before VGG16 classification

Links

- Kaggle curated lung segmentation collection:
<https://www.kaggle.com/datasets/iamtapendu/chest-x-ray-lungs-segmentation>

B2. CRD: Chest X-ray Images with Lung Segmented Masks

Best for: mask-based ROI extraction and preprocessing experiments

Modality

- Chest X-ray

Key features

- **3,311 chest X-ray images**
- Corresponding **lung segmentation masks**
- Useful for image-mask paired preprocessing and ROI extraction workflows

Why it is useful

- Good for experimenting with:
 - lung cropping
 - segmentation refinement
 - mask-guided classification
 - morphological post-processing

Link

- Kaggle dataset: <https://www.kaggle.com/datasets/mrunalInshah/crd-chest-x-ray-images-with-lung-segmented-masks>

C. Breast Cancer / Mammography Datasets

(Best if the project should remain strongly cancer-focused with lesion ROI)

These are the best choices if the project direction is **cancer-specific** rather than general chest disease intelligence.

C1. CBIS-DDSM

Best for: breast cancer detection + lesion ROI + segmentation-heavy pipeline

Modality

- Mammography

Why this dataset is important

- Strong cancer-focused dataset
- Suitable for lesion analysis, ROI extraction, and classification

Key features

- Mammography images
- Lesion annotations
- Strong support for segmentation / lesion-focused analysis

Why it is useful for the project

- Excellent if the project becomes:
 - breast cancer detection
 - lesion segmentation
 - ROI-guided deep learning
- Strong dataset for image processing + classification + lesion localization

Link

- Official TCIA page: <https://www.cancerimagingarchive.net/collection/cbis-ddsm/>

C2. INbreast

Best for: breast lesion analysis in mammography

Modality

- Full-field digital mammography

Why it is useful

- Useful for mammography-based cancer research
- Supports lesion-focused imaging tasks

Link

- Dataset information page: https://medicalresearch.inescporto.pt/breastresearch/index.php/Get_INbreast_Database

D. Brain Tumor / MRI Datasets

(Best if the project is repositioned as a segmentation-heavy MRI research project)

D1. BraTS

Best for: brain tumor segmentation + MRI-based ROI pipeline

Modality

- Brain MRI

Why this dataset is important

- One of the most recognized datasets in brain tumor segmentation research
- Strong if the project becomes a serious MRI-based tumor intelligence platform

Key features

- Brain MRI scans
- Tumor segmentation masks
- Multi-modal MRI data
- Strong research relevance

Why it is useful

- Very good for:
 - segmentation
 - ROI extraction
 - tumor visualization
 - progression-oriented MRI studies

Links

- BraTS challenge/data page: <https://www.med.upenn.edu/cbica/brats2020/data.html>
- BraTS overview: <https://www.med.upenn.edu/cbica/brats2020/>

D2. Figshare Brain MRI Tumor Dataset

Best for: simpler brain tumor starter experiments

Modality

- Brain MRI

Why it is useful

- Easier than BraTS for a first implementation

- Can be used for brain tumor classification demos

Link

- Dataset page: https://figshare.com/articles/dataset/brain_tumor_dataset/1512427

E. Lung Cancer / CT Datasets

(Best if the project must specifically target lung cancer rather than general chest abnormalities)

E1. LIDC-IDRI

Best for: lung nodule / lung cancer-oriented CT analysis

Modality

- CT scan

Why this dataset is important

- Strong option for true lung nodule / lung cancer analysis
- Radiologist-annotated nodules

Key features

- Lung CT scans
- Nodule annotations
- Suitable for ROI-driven lung cancer analysis

Link

- Official TCIA page: <https://www.cancerimagingarchive.net/collection/lidc-idri/>

E2. LUNA16

Best for: lung nodule detection benchmark

Modality

- CT scan

Why it is useful

- Derived from LIDC-IDRI
- Common benchmark for pulmonary nodule detection

Link

- Official challenge page: <https://luna16.grand-challenge.org/>

F. Histopathology / Tissue Cancer Datasets

(Best if the project should focus on microscopic tissue-level cancer detection rather than radiology)

F1. BreakHis

Best for: breast cancer histopathology classification

Modality

- Histopathology microscopy images

Why it is useful

- Useful for benign vs malignant tissue classification
- Strong if the project is tissue-image based rather than radiology based

Link

- Dataset page: <https://web.inf.ufpr.br/vri/databases/breast-cancer-histopathological-database-breakhis/>

F2. PatchCamelyon (PCam)

Best for: binary histopathology cancer classification

Modality

- Histopathology patches

Why it is useful

- Good for binary cancer classification on pathology patches
- Useful for deep learning classification workflows

Link

- Official page: <https://github.com/basveeling/pcam>
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